Faculty of Engineering

Machine Learning for Bubble Analysis in Water Electrolysers

Final Research Report submitted for EGH490-2 Semester 2 2026

Student Name:	Allen Saji (N11310375)
Supervisor:	Dr Simon Denman and Assoc Prof Dezso Sera
Submission Date:	25 October 2026

Abstract

Bubbles that form on electrolyser electrodes block active area and affect hydrogen production, but they are difficult to measure in high-speed video because they are small, densely packed and often overlapping. This project evaluated how accurately instance segmentation can delineate individual bubbles in electrolyser imagery, and whether its output supports bubble size and distribution measurements. A reproducible pipeline was built on one Semester 2 recording: frames were reconstructed from the video, the three annotators' CVAT exports were audited, and 300 annotated frames (37,388 bubble labels) were split by sequence into training, validation and test sets of 200, 50 and 50 frames. Mask R-CNN (Detectron2) was fine-tuned at three input resolutions, and StarDist and a hybrid of the two models were evaluated under the same protocol, with every model tested once on the held-out set.

Mask R-CNN with images enlarged 1.5 times reached a test mask AP of 45.1 ± 0.2 over three seeds (F1 = 0.80), compared with 40.4 at the default resolution and 43.5 for StarDist. Bubbles larger than about 25 px were found 94-100% of the time, overlapping bubbles were segmented as reliably as isolated ones, and matched diameters had a median error of 4.7%. Gas coverage (7.19% predicted versus 7.11% labelled) and the Sauter mean diameter (within 3%) were therefore accurate. No model detected bubbles smaller than about 15 px, so bubble counts were about 20% low and the mean diameter about 20% high.

On ten frames labelled by all three annotators, only 20% of sub-10 px bubbles labelled by one annotator were labelled by another, against about 96% at 25-40 px, and the model agreed with each annotator (F1 ≈ 0.75) more closely than the annotators agreed with each other (F1 ≈ 0.68). Mask R-CNN is therefore suitable for measuring electrolyser bubble coverage and size above about 20 px, and further progress on small bubbles depends mainly on consistent annotation and imaging resolution rather than on the choice of segmentation architecture.

Table of Contents

Abstract i
1.0 Introduction 1
1.1 Background and Motivation 1
1.2 Project Scope and Changes Since EGH490-1 1
1.3 Aim and Objectives 1
1.4 Research Questions 2
2.0 Literature Review 2
2.1 Classical and Detection-Based Methods 2
2.2 Segmentation Architectures 2
2.3 Electrolyser-Specific Work 3
2.4 Research Gap 3
3.0 Methodology 3
3.1 Outline 3
3.2 Dataset and Annotation 3
3.3 Frame Reconstruction and Alignment 4
3.4 Annotation Cleaning and Audit 4
3.5 Data Split 4
3.6 Mask R-CNN Baseline 5
3.7 Resolution Study and Seed Repeats 6
3.8 StarDist and Hybrid 6
3.9 Evaluation Protocol and Assumptions 6
3.10 Computing Environment 7
4.0 Results 7
4.1 Training Behaviour 7
4.2 Overall Performance 7
4.3 Performance by Bubble Size 8
4.4 Run-to-Run Variability 9
4.5 StarDist and the Hybrid 10
4.6 Overlapping Bubbles 10
4.7 Temporal Consistency 10
4.8 Reliability of Bubble Measurements 11
4.9 Inter-Annotator Agreement 12
5.0 Discussion 13
5.1 Answering the Research Questions 13
5.2 Relationship to Previous Work 13
5.3 The Small-Bubble Limit: Data Rather Than Architecture 14
5.4 Pre-processing and Background Subtraction 14
5.5 Limitations 16
6.0 Engineering and Professional Considerations 16
6.1 Risk Management 16
6.2 Ethical Considerations 17
6.3 Sustainability Considerations 17
6.4 Stakeholder Engagement 18
7.0 Potential Improvements and Future Work 18
8.0 Conclusion 18
Acknowledgements 19
Declaration of Generative AI Use 19
References 19
Appendix A: Research Project Timeline 21
Appendix B: Validation Results 22
Appendix C: Persistence Between Sequences 22
Appendix D: Frame Alignment Check 22
Appendix E: Reproducibility 23

1.0 Introduction

1.1 Background and Motivation

Water electrolysis splits water into hydrogen and oxygen at electrode surfaces and is a viable pathway to clean hydrogen when powered by renewable electricity. During electrolysis, gas bubbles nucleate, grow and detach from the electrode. While attached, they block part of the active area and alter mass transfer, so their size and spatial distribution directly influence electrolyser performance. Understanding how gas accumulates at the electrode therefore requires accurate measurement of bubble behaviour from experimental high-speed imagery.

Manual measurement of high-speed video is impractical at scale. Electrolyser bubble fields present specific difficulties: bubbles are often only a few pixels across, they sit against a textured electrode mesh, and dense packing makes individual boundaries unclear. Classical image processing such as thresholding and edge detection is sensitive to noise and lighting and degrades under dense bubble conditions (Chen & Zeng, 2024). Instance segmentation, which assigns a separate pixel mask to every object, is better suited to extracting bubble morphology, because it can separate touching bubbles and measure each one individually.

1.2 Project Scope and Changes Since EGH490-1

The Semester 1 proposal planned to compare Mask R-CNN and StarDist on the project's annotated high-speed video. Following supervisor consultation, Mask R-CNN via Detectron2 was confirmed as the primary baseline and StarDist was made conditional on available time. During Semester 2 the scope was refined further:

- **Semester 2 recording used for experiments.** Annotation of an earlier recording in Semester 1 established the CVAT workflow and the shared label schema. All reported experiments use the Semester 2 recording AutoDL_00090.mp4, whose annotations cover all three annotators and include a reference set labelled by everyone; the Semester 1 annotations are kept separate for future work.
- **Frames reconstructed from video.** CVAT image export was not available, so the annotated frames were rebuilt directly from the original video and their alignment with the annotations was verified before training.
- **StarDist completed.** Despite being conditional, StarDist was trained and evaluated under the same protocol as Mask R-CNN, together with a hybrid of the two models.
- **Additional analyses.** A resolution study, seed repeats, and analyses of temporal consistency, overlapping bubbles, measurement bias and inter-annotator agreement were added to answer the research questions more completely.

1.3 Aim and Objectives

The aim of this project is to develop and evaluate an instance segmentation approach that accurately delineates individual bubbles in high-speed electrolyser video, and to determine how reliably its output supports bubble size and distribution measurements. The objectives that support this aim are to:

- Build a verified, reproducible dataset from the Semester 2 recording and the three annotators' CVAT exports.
- Fine-tune and evaluate a Mask R-CNN baseline with a held-out test set that is used only once.

- Test whether higher input resolution improves small-bubble performance and quantify run-to-run variability.
- Evaluate StarDist under the same protocol and test whether combining the two models helps small bubbles.
- Analyse failures by bubble size, overlap and time, measure the reliability of derived bubble measurements, and quantify inter-annotator agreement.

1.4 Research Questions

The project is guided by the following primary question:

> *How accurately can a fine-tuned instance segmentation model (Mask R-CNN, compared with StarDist) delineate individual bubbles in water electrolyser high-speed video, and to what extent can its output support bubble size and distribution measurement?*

Four supporting questions are considered:

Q1. How do the models perform on small and overlapping bubbles specifically?

Q2. What practical limitations arise when training on a constrained, application-specific dataset, including input resolution, computing resources and annotation quality?

Q3. To what extent do the segmentation outputs support reliable extraction of bubble count, size and coverage?

Q4. How consistent are the human reference annotations, and how does this affect the evaluation of the models?

2.0 Literature Review

This section reviews classical and learning-based approaches to bubble detection and segmentation, the segmentation architectures evaluated in this project, and previous work on electrolyser imagery. It establishes the basis for the methods used and identifies the research gap the project addresses.

2.1 Classical and Detection-Based Methods

Early bubble detection relied on optical flow, frame differencing and edge detection, which depend on hand-crafted thresholds and fail at high bubble density (Chen & Zeng, 2024). YOLO-based detectors have been applied to electrolyser imagery; Sinapan et al. (2023) used YOLOv7 on oxygen bubbles in a PEM electrolyser and reported about 70% mAP, reflecting the difficulty of the imaging environment. Chen and Zeng (2024) improved YOLOv8n for small bubbles with deformable convolution and attention, although on a general bubble dataset. Although YOLO is often described as a bounding-box detector, segmentation variants also produce instance masks (Jocher et al., 2023; Nizovtseva et al., 2024). The relevant distinction is therefore that bounding boxes alone are insufficient for morphology measurement, so a mask-producing model is required.

2.2 Segmentation Architectures

Hessenkemper et al. (2022) compared U-Net variants with StarDist and Mask R-CNN on bubbly flow images and found StarDist and Mask R-CNN more robust under difficult imaging conditions, although deeply occluded bubbles remained unresolved by all methods. Mask R-CNN extends a two-stage detector with a mask branch on top of a ResNet feature pyramid (He et al., 2017; Lin et al., 2017) and is available with COCO-pretrained weights in Detectron2 (Wu et al., 2019). StarDist

represents each object as a star-convex polygon predicted at every pixel without anchor boxes, a design suited to small, round, densely packed objects (Schmidt et al., 2018).

2.3 Electrolyser-Specific Work

The closest electrolyser study, Pereira et al. (2026), compared U-Net and Ilastik on hydrogen bubbles from alkaline electrolysis with only 24 training images and found that both struggled with small bubbles and variable lighting. Across the reviewed literature, small-bubble detection and closely packed or overlapping bubbles were the two recurring technical challenges, compounded by small training sets.

2.4 Research Gap

The reviewed studies leave three gaps that this project addresses. No study had evaluated Mask R-CNN or StarDist on electrolyser bubble imagery or compared them on a single electrolyser dataset under consistent annotation and evaluation conditions. Small-bubble performance is usually reported as a single aggregate metric, without separating model failure from the reliability of the human labels at small sizes. Few studies test whether segmentation output supports the downstream measurements (coverage, size distribution and counts) that motivate the work. This project addresses these gaps by evaluating both architectures on the same electrolyser recording, reporting performance as a function of bubble size, measuring how well the output reproduces size and coverage, and quantifying agreement between three independent annotators.

3.0 Methodology

3.1 Outline

The project followed an end-to-end pipeline adapted from the research group's existing bubble analysis workflow (Figure 1). Data preparation was completed on a desktop computer, and model training and evaluation were run on the QUT JupyterLab platform with an NVIDIA L40S GPU (16 GB). All code, configurations, split files and logs are kept in the project folder, and every run writes its configuration, metrics and predictions to its own output folder, so each reported number can be traced to the files that produced it. The risk, ethics and sustainability considerations that shaped these choices are described in Section 6.

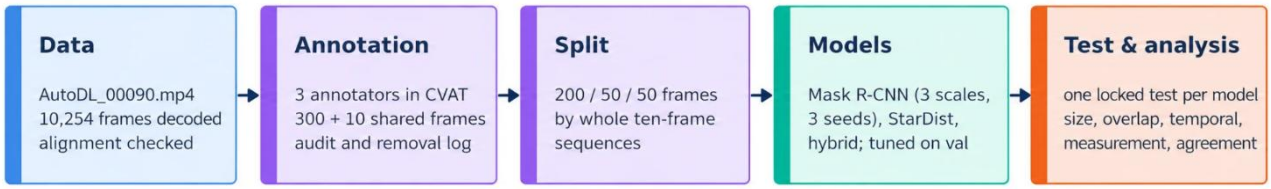


Figure 1: End-to-end pipeline from the high-speed video to the final analyses.

3.2 Dataset and Annotation

The dataset is the Semester 2 high-speed recording AutoDL_00090.mp4 of bubble formation on an electrode mesh. The file contains 10,254 frames of 1152 × 1080 pixels; the recording was captured at approximately 2,000 frames per second, so consecutive frames are about 0.5 ms apart. Three annotators, the author and two other students on the research project, each labelled ten sequences of ten consecutive frames with polygon masks in CVAT, giving 300 main frames. In this report they are referred to as Annotators 1, 2 and 3, numbered in the order in which their frames appear in the recording. In addition, all three annotators independently labelled the same ten shared frames

(5130-5139) to provide a reference for inter-annotator agreement. Annotations were exported in COCO 1.0 format.

3.3 Frame Reconstruction and Alignment

Because image export from CVAT was unavailable, the frames were reconstructed from the video. The video was decoded sequentially rather than by seeking, which avoids frame-index errors with the HEVC codec, and only the assigned frames were saved as original-resolution PNG images named with the video and frame number. Alignment between CVAT frame numbers and decoded frames was verified before bulk extraction: every export listed exactly 10,254 frames numbered from zero, matching the decoded frame count and resolution, and the annotation outlines lay on the bubbles in the decoded frames. Because the bubbles are almost static over a few frames, alignment was additionally confirmed to within one frame by measuring how well the masks fitted the foreground of each candidate frame (Appendix D).

3.4 Annotation Cleaning and Audit

The CVAT exports contained more than the assigned frames. CVAT tracks that were never closed are copied forward to every later frame, so some exports contained thousands of extra frames with stale outlines. Each export was therefore filtered to its annotator's exact frame allocation, using frame numbers recovered from the image file names rather than COCO image identifiers. Within the allocated frames, 70 labels from one export were removed because they belonged to tracks that had been drawn in an earlier sequence and carried forward without ever being edited in the current one, and one degenerate zero-area polygon was removed. Suspected duplicates, overlapping outlines and untracked shapes (31 labels) were flagged but kept. Every removal is recorded in a removal log, and the original exports were never modified (Table 1).

Table 1: Annotation audit of the three main exports (labels on the shared frames are held separately).

Annotator	Labels in export	Labels kept	Outside allocation (removed)	Carried-forward (removed)	Other removed	Flagged (kept)
Annotator 1	15,867	14,517	1,350	0	0	2
Annotator 2	14,981	12,444	2,467	70	0	11
Annotator 3	11,596	10,427	1,168	0	1	18
Total	42,444	37,388	4,985	70	1	31

Most labels inside each ten-frame sequence are CVAT copies or interpolations of a few hand-drawn keyframes (13-18% of retained labels are keyframes), and the annotators applied different thresholds for labelling faint and very small bubbles. These characteristics are analysed in Section 4.9 and are an important limitation of the reference labels.

3.5 Data Split

Frames within a ten-frame sequence are nearly identical, so whole sequences were kept together. Bubbles also persist on the same nucleation sites for a long time relative to the sequence spacing: about 52% of labelled bubbles were still present at the same location one sequence later (171 frames, about 0.09 s at 2,000 fps), and 23% nine sequences later (Appendix C). No split within a single recording can remove this correlation entirely, so held-out sequences were grouped into one contiguous block in the middle of each annotator's range, minimising the boundaries between

training and held-out data while keeping all three annotators in every split (Figure 2, Table 2). The split was fixed before any training and never changed.

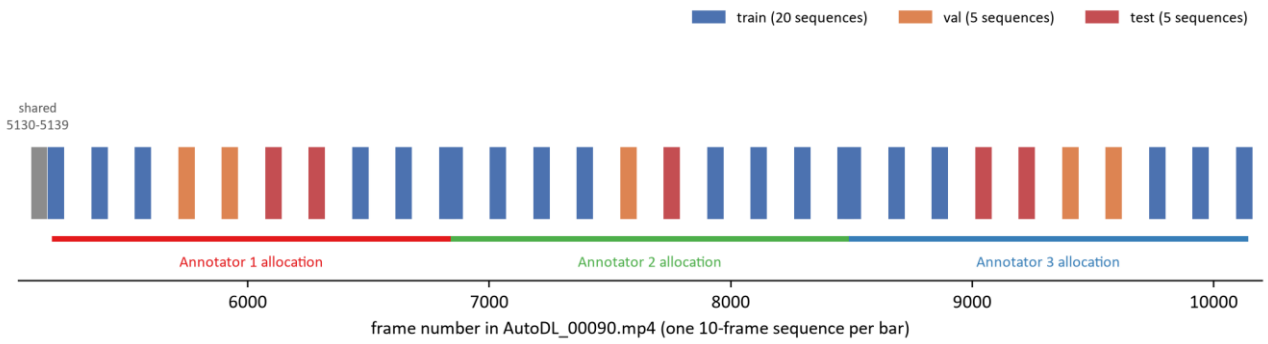


Figure 2: Allocation of the 30 ten-frame sequences to training, validation and test sets along the recording. The shared frames 5130-5139 are not in any split.

Table 2: Frozen data split.

Split	Frames	Sequences	Labels	Frames by annotator (1 / 2 / 3)
Train	200	20	25,332	60 / 80 / 60
Validation	50	5	5,898	20 / 10 / 20
Test	50	5	6,158	20 / 10 / 20

The median bubble in the training labels has an equivalent diameter of 29.7 px ($d = \sqrt{(4A/\pi)}$ for mask area A), and a quarter of labels are smaller than 20.1 px. A *small bubble* was therefore defined, before any test evaluation, as one with an equivalent diameter below 20.06 px (the training 25th percentile). Notably, 73% of all labels fall below the COCO definition of a small object (32^2 px area).

3.6 Mask R-CNN Baseline

Mask R-CNN with a ResNet-50 feature pyramid backbone was fine-tuned in Detectron2 from COCO-pretrained weights (Lin et al., 2014) for a single *bubble* class (Table 3). Because frames contain up to 159 labelled bubbles, the detection limit was raised from the COCO default of 100 to 300 per image for both the model and the primary evaluation; metrics at the conventional limit of 100 are reported alongside. A short training run on eight images confirmed that the data, masks and training loop worked before the full runs.

Table 3: Mask R-CNN training configuration.

Setting	Value
Architecture	Mask R-CNN, ResNet-50-FPN, COCO-pretrained (Detectron2 model zoo, 3x schedule weights)
Training	3,000 iterations, batch size 2, learning rate 0.0025, 200-iteration warm-up, ×0.1 decays at 70% and 90%
Input size (baseline)	Detectron2 default: shorter side 640-800 px for training, 800 px for testing (about 0.74× native)
Augmentation	Random horizontal flip and multi-scale resizing (default)
Detections per image	300 (COCO default 100 also reported)
Validation	Mask AP on the validation set every 250 iterations; best checkpoint kept
Seed	Three random seeds (one per run)

3.7 Resolution Study and Seed Repeats

The baseline shrinks each frame to about 0.74 of its native size, which turns a 10 px bubble into a 7 px object, far below the smallest anchor box of 32 px. To test whether resolution limits small-bubble detection, two further runs enlarged every input size by a constant factor: 1.5× native (shorter side 1,296-1,620 px) and 2.0× native (1,728-2,160 px). All other settings were identical. The 2.0× run exceeded the 16 GB GPU memory at full precision, so it used mixed-precision training, which leaves the schedule unchanged. The 1.5× configuration was then repeated with two additional random seeds to measure run-to-run variability. Before any test evaluation, 1.5× was fixed as the primary model because 2.0× was not clearly better on validation.

3.8 StarDist and Hybrid

StarDist 2D was trained from scratch on the same training frames, in a separate software environment (TensorFlow 2.17, StarDist 0.9.2). Overlapping labels cannot be represented in StarDist's label images, so labels were painted from largest to smallest, which kept every bubble; no labels were lost to overlap. The network used 32 rays and 256 × 256 training patches for 100 epochs of 100 steps. Two variants were trained: the standard prediction grid of 2 × 2 pixels, and a full-resolution 1 × 1 grid intended to help tiny bubbles. StarDist's own threshold optimisation was run on the validation set, and grid 1 was selected as the StarDist model before testing.

The hybrid combined the 1.5× Mask R-CNN output with StarDist detections that were smaller than a size limit and did not overlap any Mask R-CNN detection (mask IoU below 0.3). The size limit and StarDist confidence threshold were chosen on the validation set (30 px and 0.5) and the rule was then applied once to the test set.

3.9 Evaluation Protocol and Assumptions

All choices (checkpoint, confidence threshold, resolution, StarDist variant and hybrid rule) were made on the validation set only. Each model was then evaluated once on the test set, enforced by a test lock that records every attempt and refuses repeat runs unless settings are identical. The following measures were reported:

- **Mask AP** (COCO AP averaged over IoU 0.50-0.95), AP50 and AP for COCO size ranges, with a 300-detection limit.
- **Precision, recall and F1** from one-to-one matching of predicted and labelled masks at mask IoU ≥ 0.5, at the confidence threshold that maximised validation F1.
- **Small-bubble recall** (equivalent diameter < 20.06 px) and recall within diameter bins.
- **Count error** per frame and measurement error for coverage, mean diameter and Sauter mean diameter $d32 = \Sigma d^3/\Sigma d^2$.

Three assumptions underlie the evaluation. First, the labels are treated as ground truth although they are inconsistent for small bubbles; this assumption is tested directly in Section 4.9, and conclusions about small bubbles are drawn with that in mind. Second, sizes are measured in pixels because no spatial calibration was available; comparisons between models are unaffected, but absolute sizes cannot be compared with other studies. Third, held-out sequences are treated as independent of training although bubbles persist between sequences (Section 3.5), so test scores may slightly overstate performance on a new recording.

3.10 Computing Environment

Training used the QUT JupyterLab L40S GPU image (Python 3.12, PyTorch 2.13 with CUDA 12.6). The image had no C++ compiler, so a compiler was installed into the user's home directory to build Detectron2 0.6 from source. StarDist was installed in an isolated virtual environment so that TensorFlow could not interfere with the PyTorch installation. Training took about 5 minutes for the baseline, 18 minutes at 1.5× and 21 minutes at 2.0× (6.9 GB and 8.7 GB of GPU memory respectively), and 3.5 minutes (grid 2) and 7 minutes (grid 1) for StarDist.

4.0 Results

4.1 Training Behaviour

All Mask R-CNN runs converged smoothly (Figure 3). Validation mask AP reached a plateau within 500-1,500 iterations and fluctuated by about ±1 AP afterwards, so the selected checkpoints were close to the plateau rather than the result of a long upward trend. The higher-resolution runs reached a higher plateau than the baseline from the first evaluation onwards.

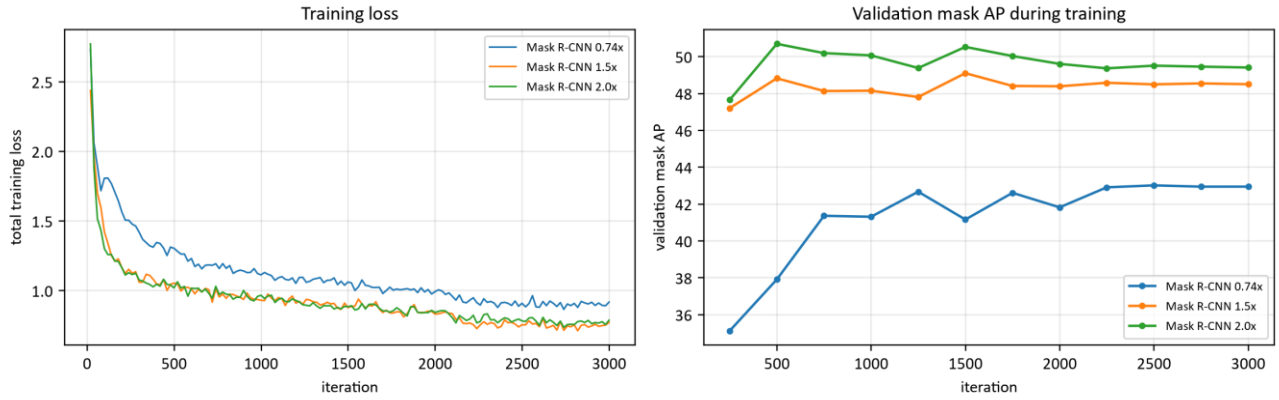


Figure 3: Training loss and validation mask AP for the three Mask R-CNN resolutions.

4.2 Overall Performance

Table 4 summarises the single official test evaluation of every model. Enlarging the input to 1.5× improved mask AP by 4.7 points and F1 by 0.03 over the baseline, and the three seeds agreed to within ±0.2 AP, so the gain is not a product of training randomness. The 2.0× model achieved the highest AP (46.7), but this came mainly from tighter masks (AP75 56.1 versus 52.3) rather than more detections: its recall, F1 and small-bubble recall were no better than 1.5×. StarDist performed between the Mask R-CNN baseline and the higher-resolution models, and the hybrid matched 1.5× Mask R-CNN without improving it.

Table 4: Official test results (50 frames, 6,158 labels). Matching at mask IoU ≥ 0.5 at each model's validation-selected threshold.

Model	Mask AP	AP50	AP75	Precision	Recall	F1	Small recall	Count error / frame
Mask R-CNN 0.74× (baseline)	40.4	65.5	47.7	0.919	0.663	0.771	0.035	−34.3
Mask R-CNN 1.5× (3 seeds)	45.1 ± 0.2	72.3	52.3	0.905	0.715	0.799	0.110	−26.0
Mask R-CNN 2.0×	46.7	74.2	56.1	0.896	0.715	0.795	0.100	−25.0
StarDist grid 2	43.2	71.6	49.2	0.925	0.666	0.775	0.037	−34.4
StarDist grid 1	43.5	70.5	49.9	0.904	0.688	0.781	0.060	−29.4
Hybrid (1.5× + StarDist)	44.5	71.0	51.8	0.888	0.725	0.798	0.112	−22.7

Validation results followed the same ranking (Appendix B). Test scores were 3-4 AP lower than validation scores for every model, which is expected because checkpoints and thresholds were chosen to maximise validation performance, and the test set contains a larger share of small bubbles (27% versus 24%).

4.3 Performance by Bubble Size

Figure 4 shows recall as a function of bubble size and is the central result of the project. All models found 94-100% of bubbles above 25 px. Between 15 and 25 px, recall depended strongly on resolution: at 15-20 px it rose from 13% (baseline) to 38% (1.5×), and at 20-25 px from 66% to 82%. Below 15 px, recall was effectively zero for every model, including StarDist and the hybrid. This floor affects a large share of the labels: 1,226 of the 6,158 test labels (20%) are smaller than 15 px.

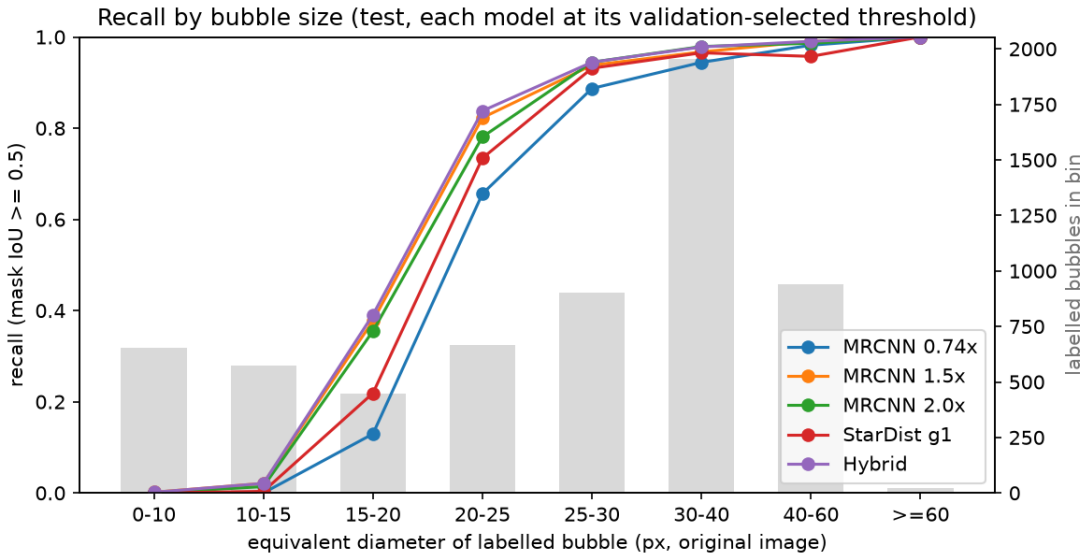


Figure 4: Test recall by bubble size for the main models. Grey bars show the number of labelled bubbles in each size bin.

The size distributions in Figure 5 make the same point from the other direction: the labels contain a distinct population of bubbles around 7-15 px, while every model's predictions begin at about 15 px.

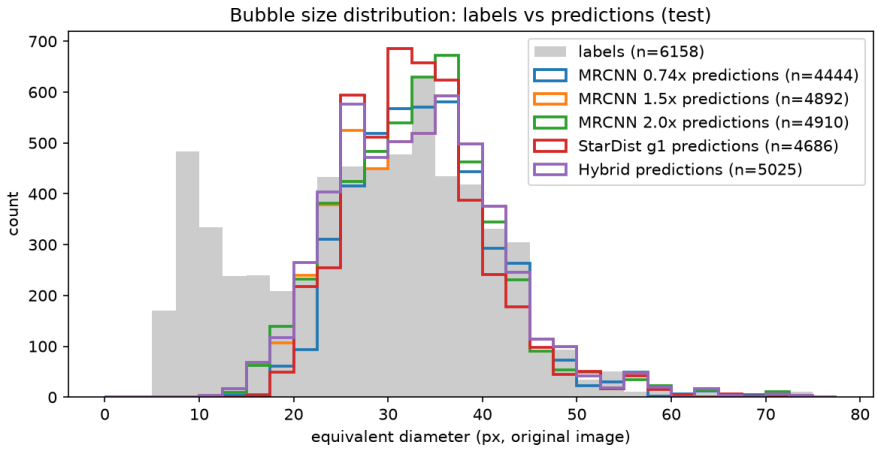


Figure 5: Size distribution of labelled bubbles and of each model's predictions on the test set.

A diagnostic on the validation set showed that the small bubbles are largely missed rather than poorly outlined. Relaxing the matching criterion from IoU 0.5 to 0.1 raised the baseline's small-bubble recall only from 1.8% to 3.9%, and lowering the confidence threshold to 0.05 raised the 2.0× model's small-bubble recall to 26%, at a large cost in precision. The models therefore detect some small bubbles, but with low confidence, so the threshold that maximises overall accuracy discards them. Figure 6 shows a typical test frame: larger bubbles are outlined closely, while several small, labelled bubbles have no matching prediction.

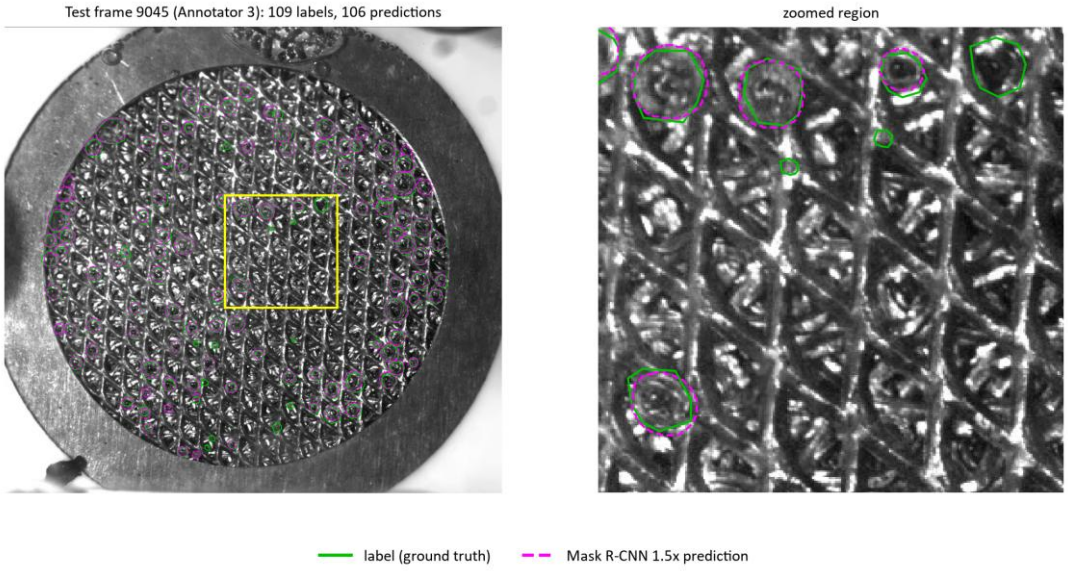


Figure 6: Mask R-CNN 1.5× predictions (magenta, dashed) and labels (green) on a test frame, with an enlarged region.

4.4 Run-to-Run Variability

Table 5 shows the 1.5× Mask R-CNN model trained with three different random seeds and otherwise identical settings.

Table 5: Variability of the 1.5× Mask R-CNN model across three random seeds.

Run	Val mask AP	Test mask AP	Test F1	Test small recall	Test count error
1	49.1	45.2	0.800	0.109	−25.3
2	49.0	45.3	0.799	0.106	−25.8
3	49.5	44.9	0.798	0.115	−26.8
Mean ± SD	49.2 ± 0.3	45.1 ± 0.2	0.799 ± 0.001	0.110 ± 0.005	−26.0 ± 0.8

Variation between seeds (±0.2 AP on test) is more than twenty times smaller than the gain from resolution, so the comparison between resolutions is robust to training randomness.

4.5 StarDist and the Hybrid

StarDist, trained from scratch without pretrained weights, reached 96% of the 1.5× Mask R-CNN mask AP on test. Predicting at full resolution (grid 1) improved small-bubble recall from 3.7% to 6.0% at a better F1, but did not remove the small-bubble floor (Figure 4). Like Mask R-CNN, StarDist assigned low confidence to small bubbles: at a confidence of 0.1 the grid 2 model recovered 17% of small validation bubbles, but precision fell to 0.64. Because StarDist rarely found small bubbles that Mask R-CNN missed, the hybrid's small gain on validation (F1 0.821 versus 0.816) disappeared on test (0.798 versus 0.800). Two architectures with entirely different designs, an anchor-based two-stage detector and an anchor-free star-convex model, therefore reached the same limit.

4.6 Overlapping Bubbles

In the test labels, 1,067 bubbles (17%) overlap another labelled bubble and 203 touch one. For bubbles of 20 px and above, recall was 0.94 for isolated, 0.97 for touching and 0.96 for overlapping bubbles with the 1.5× model, and the IoU of matched masks was 0.82-0.84 in every case (Figure 7). Overlap therefore did not limit performance; size did.

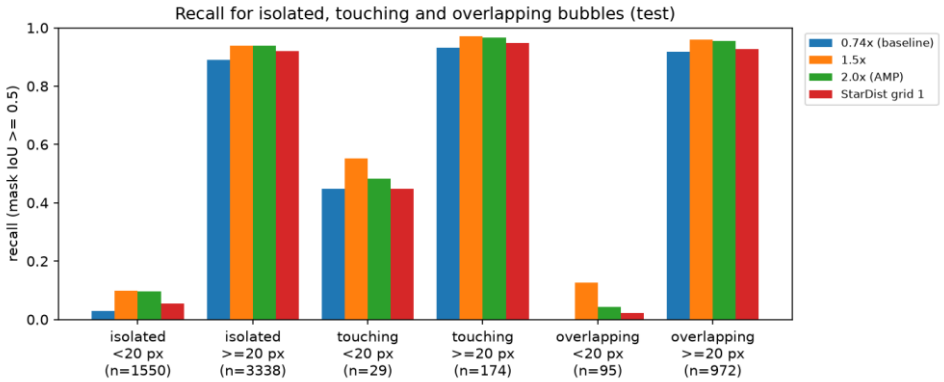


Figure 7: Test recall for isolated, touching and overlapping bubbles, split at 20 px.

4.7 Temporal Consistency

Because each test sequence contains ten consecutive frames and the labels track each bubble through the sequence, it is possible to check whether a missed bubble is missed consistently or intermittently. Figure 8 shows that bubbles above 25 px were found in every frame of their sequence 86-100% of the time, and bubbles below 15 px were never found in 88-98% of cases. Intermittent detection was concentrated at 15-20 px, where 53% of bubbles were found in only some frames. At this size, 69% of bubbles were detected in at least one frame although only 38% were detected in any single frame, so linking detections across frames could recover some near-threshold bubbles, but temporal interpolation cannot help below 15 px.

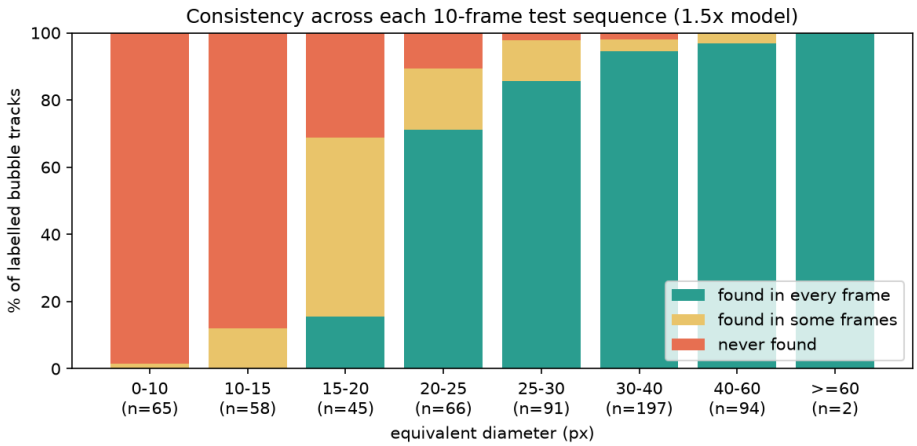


Figure 8: Whether each labelled bubble was found in every frame, some frames or no frame of its ten-frame test sequence (1.5× model).

4.8 Reliability of Bubble Measurements

Table 6 compares measurements derived from the 1.5× model with the same measurements derived from the labels. When a bubble was found, its diameter was reproduced closely (median absolute error 4.7%, mean bias +1.5%). Area-weighted measurements were therefore reliable: predicted gas coverage was 7.19% of the frame against 7.11% labelled, and the Sauter Mean Diameter differed by 3%. Number-weighted measurements were biased because the missed bubbles are numerous but small: the bubble count was 20% low and the mean diameter 20% high.

Table 6: Bubble measurements from the model compared with the labels.

Measurement (test set)	From labels	From model (1.5×)	Difference
Diameter of a found bubble	n/a	median |error| 4.7%	bias +1.5%
Gas coverage (% of frame)	7.11	7.19	+1%
Sauter Mean Diameter d32 (px)	36.6	37.9	+3%
Bubbles per frame	123.2	98.2	−20%
Mean equivalent diameter (px)	27.9	33.4	+20%

Figure 9 repeats the comparison for each test sequence. The labelled bubble count depends strongly on the annotator: about 143 bubbles per frame in Annotator 1's sequences, 126 in Annotator 2's and 96-108 in Annotator 3's, while the model's count stayed between 90 and 106. The count error was correspondingly small on Annotator 3's frames (2.2 bubbles per frame) and large on Annotator 1's (50.2). This reflects how many faint bubbles each annotator chose to label rather than differences in model accuracy, as examined next.

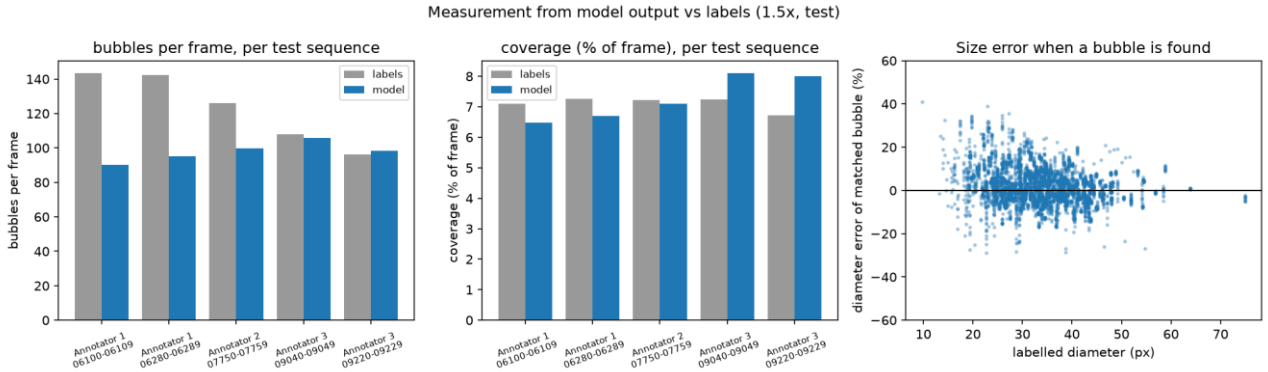


Figure 9: Bubble count and coverage per test sequence (labels versus model), and diameter error of matched bubbles.

4.9 Inter-Annotator Agreement

On the ten shared frames, the three annotators labelled 1,350 (Annotator 1), 1,277 (Annotator 2) and 1,056 (Annotator 3) bubbles on identical images, a 28% difference in count. Figure 10 overlays the three annotations on one frame. Large, clear bubbles are outlined by all three annotators with only small differences in boundary placement, whereas many faint and small bubbles were labelled by only one person.

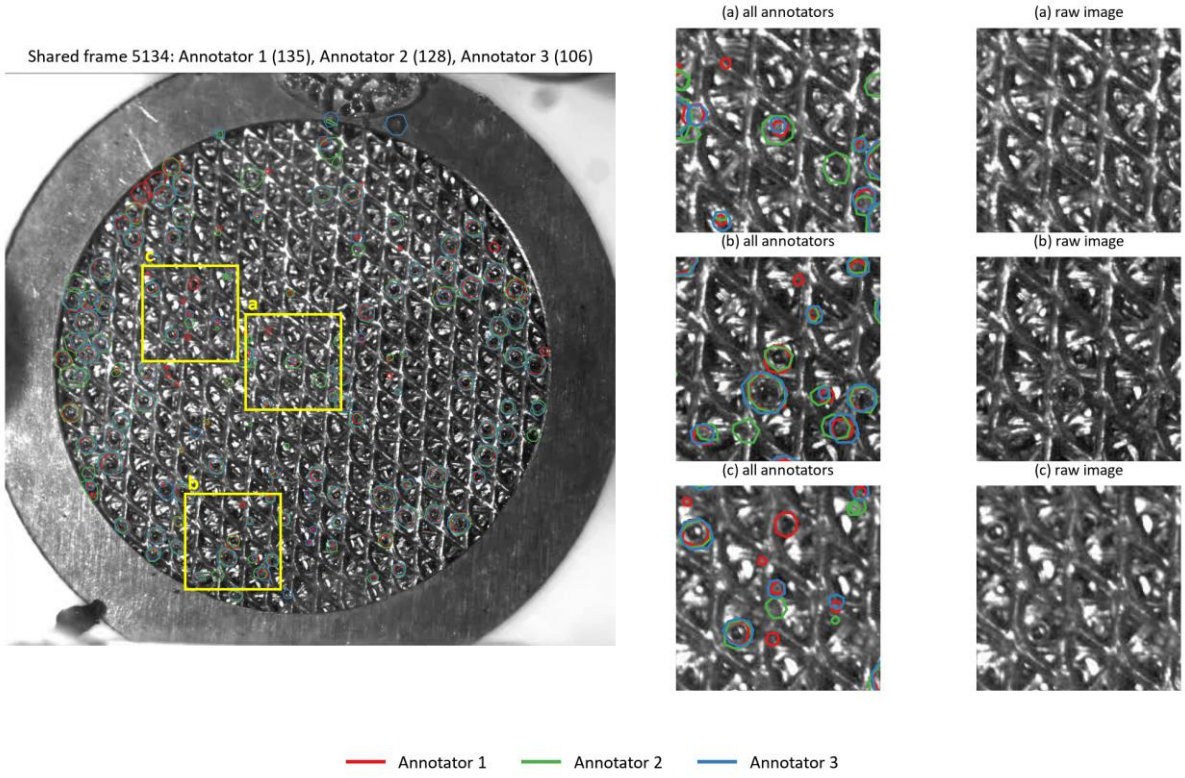


Figure 10: Independent annotations of shared frame 5134 by Annotator 1 (red), Annotator 2 (green) and Annotator 3 (blue), with enlarged regions where the annotators disagree.

Using the same one-to-one matching as for the models, the annotators agreed with each other at F1 0.665-0.687, while the 1.5× model agreed with each annotator at F1 0.745-0.756 (Table 7). Agreement depended strongly on size (Figure 11): only 20% of sub-10 px bubbles labelled by one annotator were also labelled by another, rising to 58% at 15-20 px and about 96% at 25-40 px. Model recall followed the same curve. The model found 93% of bubbles that all three annotators labelled (98% for those above 20 px) but only 5% of bubbles labelled by a single annotator; StarDist behaved the same way (88% and 4%).

Table 7: Agreement between annotators, and between the model and each annotator. For annotator pairs F1 = 2 × matched / (labels of both).

Comparison (shared frames, IoU ≥ 0.5)	Agreement (F1)
Annotator 1 vs Annotator 2	0.665
Annotator 1 vs Annotator 3	0.687
Annotator 2 vs Annotator 3	0.679
Mask R-CNN 1.5× vs Annotator 1	0.745
Mask R-CNN 1.5× vs Annotator 2	0.749
Mask R-CNN 1.5× vs Annotator 3	0.756

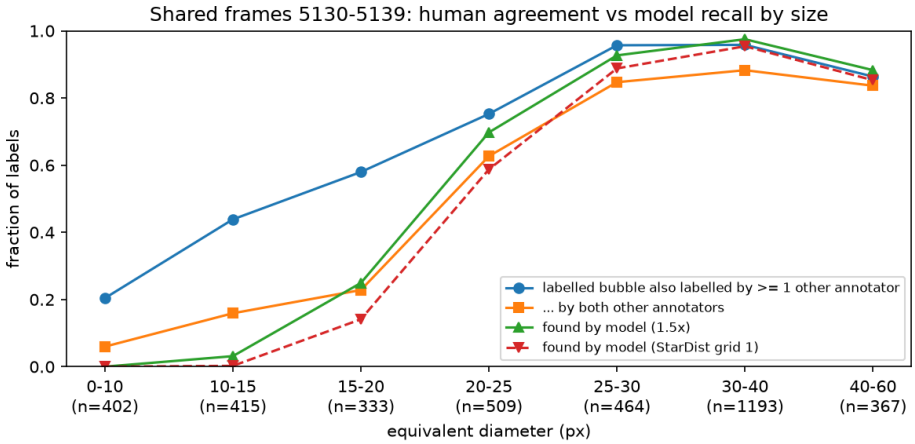


Figure 11: Fraction of each annotator's labels confirmed by the other annotators, and fraction found by each model, by bubble size (shared frames).

The shared frames are only 61 frames before the first training sequence and represent a single moment of the recording, so these comparisons are strong evidence rather than proof. They nevertheless show that the human reference itself becomes unreliable at the sizes where the models fail.

5.0 Discussion

5.1 Answering the Research Questions

Primary question. A fine-tuned Mask R-CNN delineates electrolyser bubbles accurately above about 20-25 px: such bubbles are found 94-100% of the time, including overlapping bubbles, and their diameters are reproduced to within about 5%. With a 1.5× input scale the model reached a test mask AP of 45.1 and F1 of 0.80. StarDist was competitive but not better. The output supports area-weighted measurements (coverage, Sauter mean diameter) but not bubble counts or number-weighted size distributions.

Small and overlapping bubbles (Q1). Overlap was not a limiting factor. Small bubbles were: recall was near zero below 15 px for every model, and partial (38%) at 15-20 px. Higher resolution moved the detection floor from about 20 px to about 15 px but no further.

Practical limitations (Q2). The main practical limits were GPU memory (2.0× required mixed precision, and larger scales were not feasible on 16 GB), the absence of a compiler on the training platform, and, most importantly, the characteristics of the labels. Resolution beyond 1.5× gave tighter masks but no additional detections.

Measurement (Q3). Coverage and Sauter diameter can be measured reliably; counts and mean diameter are biased by about 20% because the missing bubbles are small. This matters for interpretation: a model like this would systematically report fewer, larger bubbles than are present.

Annotation consistency (Q4). Annotators disagreed substantially on small and faint bubbles, and the model's failures coincide with those disagreements.

5.2 Relationship to Previous Work

The results address each gap identified in Section 2.4. This is the first evaluation of Mask R-CNN and StarDist on the same electrolyser recording under one protocol among the studies reviewed. Consistent with Hessenkemper et al. (2022), who found both models robust on difficult bubbly-flow

images, the two architectures performed similarly here, and overlapping bubbles were segmented as reliably as isolated ones. The small-bubble difficulty reported by Chen and Zeng (2024) and Pereira et al. (2026) was also observed, but the size-resolved and inter-annotator analyses add an explanation that earlier studies could not separate: below about 15 px the human labels themselves are inconsistent. The mask AP values are not directly comparable with the 70% mAP reported for YOLOv7 by Sinapan et al. (2023), which is a bounding-box detection metric on a different dataset. Instead, the measurement analysis shows which derived quantities a mask-based model delivers reliably (coverage and Sauter diameter) and which it does not (counts and number-weighted means).

5.3 The Small-Bubble Limit: Data Rather Than Architecture

Several independent observations indicate that the small-bubble limit is caused mainly by the data rather than by the choice of model. First, two architectures with fundamentally different designs reached the same floor; StarDist has no anchor boxes and predicts at every pixel, yet it did not find sub-15 px bubbles either. Second, both models detected some small bubbles but assigned them low confidence, which is the expected behaviour when the training labels are inconsistent: the same kind of faint, few-pixel object is labelled as a bubble in some frames and left as background in others. Third, the floor coincides with the size at which human annotators stop agreeing. In Semester 1 the progress report identified annotation consistency across contributors as a risk, noting that small bubbles require judgement calls that may differ between annotators; the results show that this risk materialised and was the main limit on small-bubble performance.

It would be incorrect to treat any single annotator as correct. A faint few-pixel feature may be a real bubble, a reflection or noise, and without an agreed labelling rule there is no ground truth at this scale. Part of the floor is also physical: at the native resolution an 8 px bubble covers only a few dozen pixels against a strongly textured mesh, so even consistent labels may not contain enough information for reliable detection.

5.4 Pre-processing and Background Subtraction

Pre-processing steps such as contrast enhancement were explored early in the project but could not be carried through: they altered the images in ways that no longer matched the existing annotations and complicated the annotation workflow, so all models were trained on unmodified frames.

Background subtraction was also explored as a classical alternative to a learned model. Because the electrode mesh is static, a temporal-median background over ±900 frames removes the mesh texture and leaves the parts of the image that change (Figure 12). The difference is on average about five times stronger inside labelled bubbles than elsewhere, so background subtraction does highlight many bubbles; most bubbles are not present for long enough within the window to be absorbed into the background.

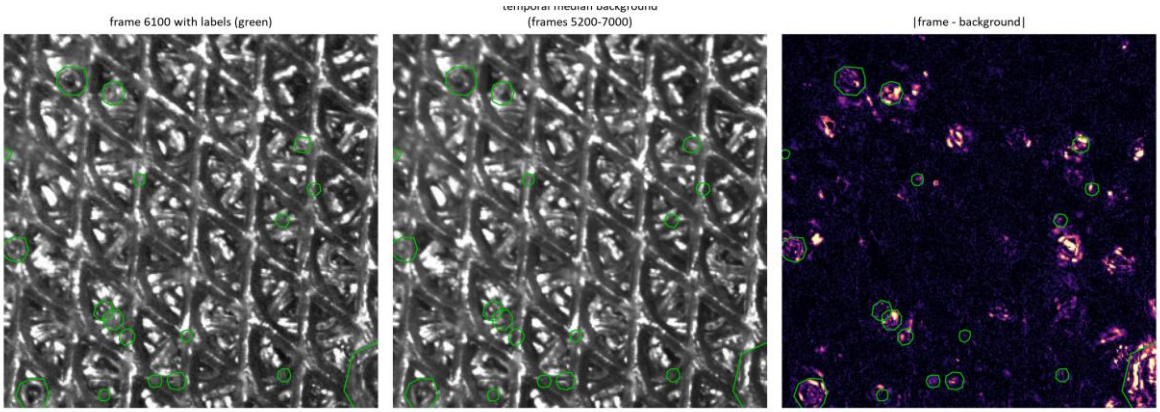


Figure 12: Background subtraction on test frame 6100, showing the frame with labels, the temporal-median background and the absolute difference.

Turning the difference image into a segmentation, however, exposes its limits (Figure 13). Thresholding the top 5% of differences within the cell gave a pixel-level Dice overlap with the labelled bubbles of only 0.31 and 0.26 on two test frames from different parts of the recording, and produced about 550 separate foreground fragments against 144 and 109 labelled bubbles. Large bubbles break into several fragments because only their edges and highlights differ from the background, other changing features such as reflections create false foreground, and faint small bubbles barely exceed the threshold. Background subtraction yields a foreground map rather than separated bubble instances, so it cannot replace instance segmentation, although it could be investigated as an additional input channel in future work.

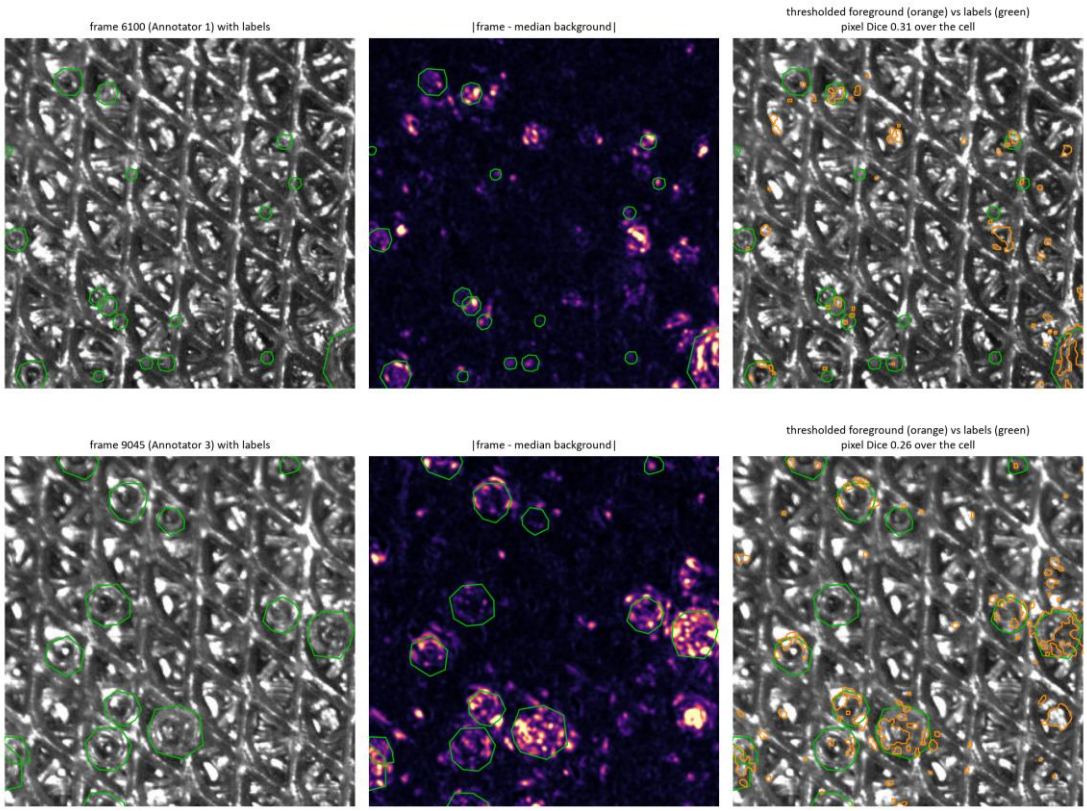


Figure 13: Threshold background-subtraction foreground (orange) compared with the labels (green) on two test frames, with the pixel Dice overlap over the cell.

5.5 Limitations

- **Single recording.** All frames come from one video, so the results describe performance within this recording. Bubbles persist across sequences, which makes held-out sequences correlated with training data; generalisation to other electrolysers, currents or lighting is untested.
- **Reference labels.** Labels are mostly CVAT interpolations of a few keyframes, include kept duplicates, and differ systematically between annotators. Each annotator also covers a different time window, so annotator style and time are confounded.
- **Small evaluation sets.** Validation and test each contain five correlated sequences; differences below about 1 AP should not be over-interpreted.
- **Physical units.** Sizes are reported in pixels because no spatial calibration was available.
- **Scope of comparison.** No YOLO segmentation model or the group's existing pipeline was evaluated on the same split, so claims are limited to Mask R-CNN and StarDist.

6.0 Engineering and Professional Considerations

This section reviews how the risks, ethical principles, sustainability objectives and stakeholders identified in EGH490-1 were managed during delivery, and what the outcome was.

6.1 Risk Management

The largest schedule risk identified in Semester 1, annotation delay, materialised and moved all model training into Semester 2 (Appendix A). Table 8 revisits the EGH490-1 risk register, using its original likelihood and overall rating (red high or likely, yellow medium or possible, green low or unlikely), and records what happened during delivery and how each risk was handled. Two risks were added in the progress report, and one new challenge that arose in Semester 2 has been rated in hindsight; all were managed so that the core deliverables were completed.

Table 8: EGH490-1 risk register revisited with Semester 2 outcomes.

Risk	Likelihood	Rating	Outcome in Semester 2	Mitigation and response
Insufficient or imbalanced annotated data	Likely	High	300 frames (37,388 labels) of one recording were sufficient for training; other recordings untested	Semester 2 recording used for all experiments; single-recording scope stated as a limitation
Annotation delays from team coordination	Likely	High	Materialised: all training moved to Semester 2, leaving less time for iteration	Baseline training made the first Semester 2 priority
Models underperform on small or overlapping bubbles	Possible	Medium	Materialised below about 15 px; overlap was not a limit	Resolution study, StarDist and hybrid tested; size-resolved reporting
Training instability during fine-tuning	Possible	Medium	Did not occur; all runs converged	COCO-pretrained weights and conservative settings
GPU or HPC access restricted	Possible	Medium	No compiler on the training image; 16 GB memory limit at 2.0×	User-space compiler build; mixed precision; isolated StarDist environment
Scope uncertainty in early stages	Possible	Medium	Resolved: StarDist completed in addition to Mask R-CNN	Mask R-CNN prioritised; StarDist added once the baseline was complete
Video quality limiting annotation and accuracy	Unlikely	Low	Planned pre-processing changed the images so they no longer matched the annotations	Unmodified frames for all models; background subtraction evaluated (Section 5.4)
Annotation consistency across contributors (added in EGH490-1 progress report)	Likely	Medium	Materialised: annotators disagreed on small and faint bubbles; main limit on small-bubble performance	Measured on shared frames; reported as a limitation; labelling rule recommended
Evaluation set too small or not representative (added in EGH490-1 progress report)	Possible	Medium	Bubbles persist between sequences, so held-out data is correlated	Split by whole sequences and fixed before training; test set used once per model
CVAT image export unavailable; unclosed tracks (new in Semester 2, rated in hindsight)	Likely	Medium	Frames had to be rebuilt; stale labels in exports	Frames decoded from video with an alignment check; audited filtering with a removal log

6.2 Ethical Considerations

The project involved no human participants or sensitive data and required no ethics approval; the video was recorded internally by the research group. All pretrained weights and external software are cited at their source, and the contributions of the other annotators are acknowledged. Because their labelling is compared directly, the annotators are referred to by number rather than by name, and no annotator is treated as correct: disagreement is reported as a property of the task. In line with Principle 1 (Demonstrate Integrity) of the Engineers Australia Code of Ethics (Engineers Australia, 2022), the test set was evaluated only once per model, every data exclusion was logged, and negative results and limitations are reported directly.

6.3 Sustainability Considerations

Transfer learning from COCO weights and short training schedules kept each training run below about 25 minutes of GPU time, and because all tuning used the validation set, no runs were repeated against the test set; this is consistent with Principle 4 (Promote Sustainability) of the Code of Ethics.

The scripts, configurations and logs (Appendix E) allow the research group to extend the work without repeating data preparation. In the longer term, better measurement of electrolyser bubble behaviour supports the development of renewable hydrogen production, in line with United Nations Sustainable Development Goal 7, Affordable and Clean Energy (United Nations, 2015).

6.4 Stakeholder Engagement

The primary stakeholders were the supervisors, Dr Simon Denman and Assoc Prof Dezso Sera, who guided the project through regular consultation across both semesters. Their input confirmed Mask R-CNN via Detectron2 as the first baseline and made StarDist conditional on time (Section 1.2). Dr Denman also gave valuable feedback on how to proceed with the project, the annotation task, and image processing and pre-processing.

The research group provided the high-speed recordings, the existing bubble analysis pipeline and supporting resources on which the workflow was based. The two other thesis students on the research project were both stakeholders and team members: the annotation was divided between the three students, so each depended on the others completing their allocations under the same label schema, and all three labelled the shared reference frames used in Section 4.9. QUT provided the GPU platform and academic oversight.

7.0 Potential Improvements and Future Work

The results point to the following next steps, in order of expected benefit:

- **Consistent small-bubble labels.** Agree a written labelling rule for faint and small bubbles (for example, a minimum size and visibility criterion) and build consensus labels from multiple annotators. This is the improvement most likely to raise small-bubble performance.
- **Higher optical resolution or tiling.** Capture at higher magnification, or train on image tiles at higher scale, so that small bubbles span more pixels.
- **Small-object model changes.** Smaller anchor sizes and a finer feature level for Mask R-CNN, or pretrained StarDist weights, are natural next experiments once labels are consistent.
- **Temporal linking.** Associating detections across consecutive frames could recover bubbles near the 15-20 px threshold that are detected intermittently.
- **Broader evaluation.** Additional recordings and operating conditions, a calibrated pixel size for physical units, and a comparison with YOLO segmentation and the group's existing pipeline on the same split.

8.0 Conclusion

This project delivered a verified dataset, a reproducible training and evaluation pipeline, and a controlled comparison of Mask R-CNN and StarDist for bubble instance segmentation in electrolyser imagery. Mask R-CNN at 1.5× input scale achieved a test mask AP of 45.1 ± 0.2 and F1 of 0.80, segmented overlapping bubbles as reliably as isolated ones, and reproduced gas coverage and Sauter Mean Diameter to within a few percent. No model detected bubbles smaller than about 15 px, the size at which the three annotators stop agreeing, and the model agreed with each annotator more closely than the annotators agreed with each other. Reliable small-bubble measurement therefore depends first on consistent annotation and adequate imaging resolution; with those in place, the pipeline provides a basis for further work by the research group.

Acknowledgements

The author thanks Dr Simon Denman and Assoc Prof Dezso Sera for their guidance throughout the project, the research group for providing the high-speed recordings, the existing bubble analysis pipeline and supporting resources, and the two fellow thesis students on the research project for their annotation work and the shared reference frames.

Declaration of Generative AI Use

The generative AI tool Claude Code (Anthropic) was used during this project to assist with code development and debugging, and with drafting and refining written content. This assistance included improving the clarity and structure of the academic presentation of the report and debugging code where Jupyter Notebook runs failed. The research methodology, experimental design, implementation decisions, and evaluation and interpretation of results were undertaken and critically reviewed by the author. All AI-assisted outputs were reviewed and verified before inclusion in the final submission.

References

Chen, T., & Zeng, Q. (2024). Research on bubble detection based on improved YOLOv8n. *IEEE Access, 12*, 9659-9672. https://doi.org/10.1109/ACCESS.2024.3353196

Engineers Australia. (2022). *Code of ethics and guidelines on professional conduct*. https://www.engineersaustralia.org.au/sites/default/files/2022-08/code-ethics-guidelines-professional-conduct-2022.pdf

He, K., Gkioxari, G., Dollár, P., & Girshick, R. (2017). Mask R-CNN. In *Proceedings of the IEEE International Conference on Computer Vision* (pp. 2961-2969). https://doi.org/10.1109/ICCV.2017.322

Hessenkemper, H., Starke, S., Atassi, Y., Ziegenhein, T., & Lucas, D. (2022). Bubble identification from images with machine learning methods. *International Journal of Multiphase Flow, 155*, Article 104169. https://doi.org/10.1016/j.ijmultiphaseflow.2022.104169

Jocher, G., Chaurasia, A., & Qiu, J. (2023). *Ultralytics YOLOv8* (Version 8.0.0) [Computer software]. https://github.com/ultralytics/ultralytics

Lin, T.-Y., Dollár, P., Girshick, R., He, K., Hariharan, B., & Belongie, S. (2017). Feature pyramid networks for object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 2117-2125). https://doi.org/10.1109/CVPR.2017.106

Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., & Zitnick, C. L. (2014). Microsoft COCO: Common objects in context. In *Computer Vision – ECCV 2014* (pp. 740-755). Springer. https://doi.org/10.1007/978-3-319-10602-1_48

Nizovtseva, I., Mikushin, P., Starodumov, I., Makhaeva, K., Kraev, S., & Chernushkin, D. (2024). Bubble detection in multiphase flows through computer vision and deep learning for applied modeling. *Mathematics, 12*(23), 3864. https://doi.org/10.3390/math12233864

Pereira, J., Souza, R., Normand, A., & Moita, A. (2026). Image-based segmentation of hydrogen bubbles in alkaline electrolysis: A comparison between Ilastik and U-Net. *Algorithms, 19*(1), 77. https://doi.org/10.3390/a19010077

Schmidt, U., Weigert, M., Broaddus, C., & Myers, G. (2018). Cell detection with star-convex polygons. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018* (pp. 265-273). Springer. https://doi.org/10.1007/978-3-030-00934-2_30

Sinapan, I., Lin-Kwong-Chon, C., Damour, C., Kadjo, J.-J. A., & Benne, M. (2023). Oxygen bubble dynamics in PEM water electrolyzers with a deep-learning-based approach. *Hydrogen, 4*(3), 556-572. https://doi.org/10.3390/hydrogen4030036

United Nations. (2015). *Transforming our world: The 2030 Agenda for Sustainable Development* (A/RES/70/1). https://sdgs.un.org/2030agenda

Wu, Y., Kirillov, A., Massa, F., Lo, W.-Y., & Girshick, R. (2019). *Detectron2* [Computer software]. https://github.com/facebookresearch/detectron2

Appendix A: Research Project Timeline

Table 9 summarises how the project was completed across the two semesters, comparing the timeline planned in EGH490-1 with what was delivered. Annotation took longer than planned, so all model training moved to Semester 2. Dataset preparation became a milestone of its own because CVAT image export was unavailable, and StarDist, planned as conditional, was completed together with the additional analyses.

Table 9: Planned and delivered milestones across Semester 1 (S1) and Semester 2 (S2).

Milestone	Planned (EGH490-1)	Delivered	Outcome
Project proposal and scope of work	S1 Week 7	S1 Week 7	Complete on schedule
Annotated dataset (CVAT)	S1 Week 8	S1 Weeks 10-13	Complete; annotation workload higher than estimated
Methodology refinement and model selection	S1 Weeks 8-9	S1 Weeks 8-9	Complete; Mask R-CNN via Detectron2 confirmed as first baseline
Initial baseline experiments	S1 Weeks 9-12	Deferred to S2	Deferred because annotation extended to the end of Semester 1
Progress report and presentation	S1 Week 13	S1 Week 13	Complete on schedule
Dataset preparation (frames, audit, split)	Not planned separately	S2 Week 9	Complete; frames rebuilt from video, exports audited, split frozen
Mask R-CNN baseline training	S2 Weeks 2-5	S2 Week 10	Complete; resolution study and seed repeats added
Model evaluation (mask AP, F1)	S2 Weeks 5-7	S2 Week 11	Complete; each model tested once
Failure case analysis	S2 Weeks 7-9	S2 Week 12	Complete; size, overlap, temporal, measurement and annotator analyses
StarDist evaluation (conditional)	S2 Week 9	S2 Week 12	Complete, together with the hybrid model
Draft thesis	S2 Weeks 10-11	S2 Week 13	Complete
Final report and oral presentation	S2 Weeks 12-13	S2 Week 13	Submitted

Appendix B: Validation Results

Table 10 lists the validation results on which every model-selection decision was based (Section 3.9).

Table 10: Validation results (50 frames, 5,898 labels), used for all model selection.

Model	Mask AP	AP50	Precision	Recall	F1	Small recall	Count error
Mask R-CNN 0.74×	43.0	67.0	0.925	0.684	0.786	0.018	−30.7
Mask R-CNN 1.5× (run 1)	49.1	75.3	0.909	0.741	0.816	0.132	−21.8
Mask R-CNN 1.5× (run 2)	49.0	74.9	0.901	0.746	0.816	0.131	−20.4
Mask R-CNN 1.5× (run 3)	49.5	75.5	0.913	0.734	0.814	0.128	−23.2
Mask R-CNN 2.0×	50.7	76.9	0.900	0.735	0.809	0.107	−21.5
StarDist grid 2	47.0	74.7	0.899	0.704	0.789	0.011	−25.6
StarDist grid 1	47.6	74.0	0.898	0.726	0.803	0.070	−22.7
Hybrid	49.0	74.4	0.887	0.765	0.821	0.146	−16.3

Appendix C: Persistence Between Sequences

Table 11 shows how long labelled bubbles stay at the same location, which is why whole sequences were kept together when the data was split (Section 3.5).

Table 11: Fraction of labelled bubbles in one frame matched (mask IoU ≥ 0.5) by a label in a later sequence.

Sequence lag	Median gap (frames)	Approx. time at 2,000 fps	Median fraction of labels still present
1	171	0.09 s	0.52
2	351	0.18 s	0.43
3	531	0.27 s	0.38
5	891	0.45 s	0.29
9	1,461	0.73 s	0.23

Appendix D: Frame Alignment Check

Every CVAT export listed exactly 10,254 frames numbered from zero, matching the decoded video. Figure 14 draws the outlines from CVAT frame 8474 on decoded frames 8472, 8474 and 8476. The outlines sit on the bubbles at the matching frame (green). Because the bubbles barely move over a few frames, neighbouring frames look almost identical, so alignment was also confirmed numerically: a score measuring how well the masks fit the changing foreground, computed for offsets of up to ±300 frames, peaked within ±1 frame in six of the seven samples tested and fell away clearly by ±10 to ±30 frames.

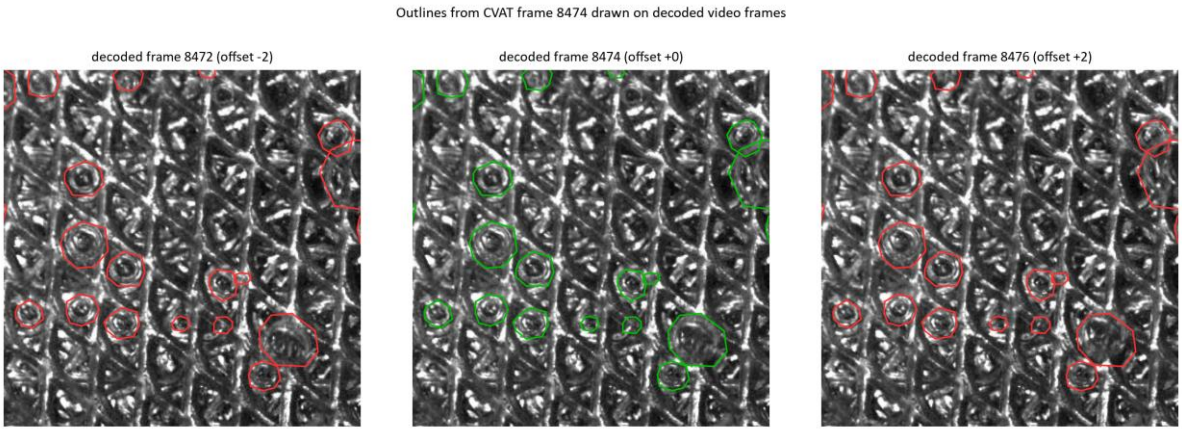


Figure 14: Frame alignment check, showing outlines from CVAT frame 8474 on decoded frames 8472, 8474 (green) and 8476.

Appendix E: Reproducibility

The code, configurations and result summaries are available at https://github.com/unholypot/egh490-bubble-segmentation. The recordings and raw annotation exports belong to the research group and are not included. The repository is organised as follows:

- **Data preparation:** scripts/check_frame_alignment.py, scripts/prepare_semester2.py, scripts/make_splits.py
- **Training and evaluation (QUT):** notebooks/02_train_and_evaluate.ipynb, notebooks/03_stardist.ipynb, configs/*.json
- **Analysis and figures:** scripts/analyze_results.py, scripts/compare_all_runs.py, scripts/hybrid_mrcnn_stardist.py, scripts/make_report_figures.py
- **Records:** annotation audit, removal log, split manifest and rationale, and each run's configuration, metrics, predictions and test-attempt log